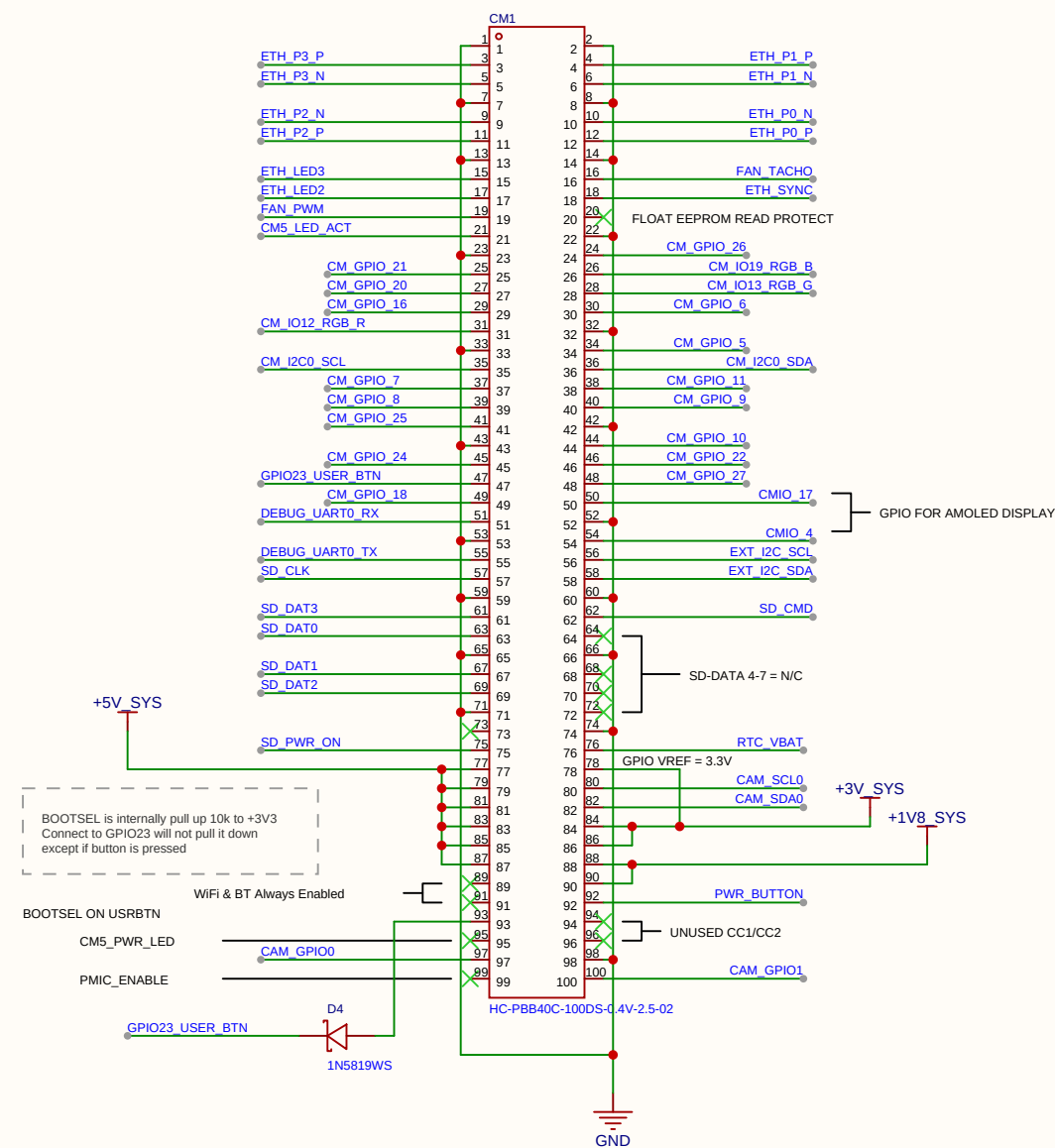
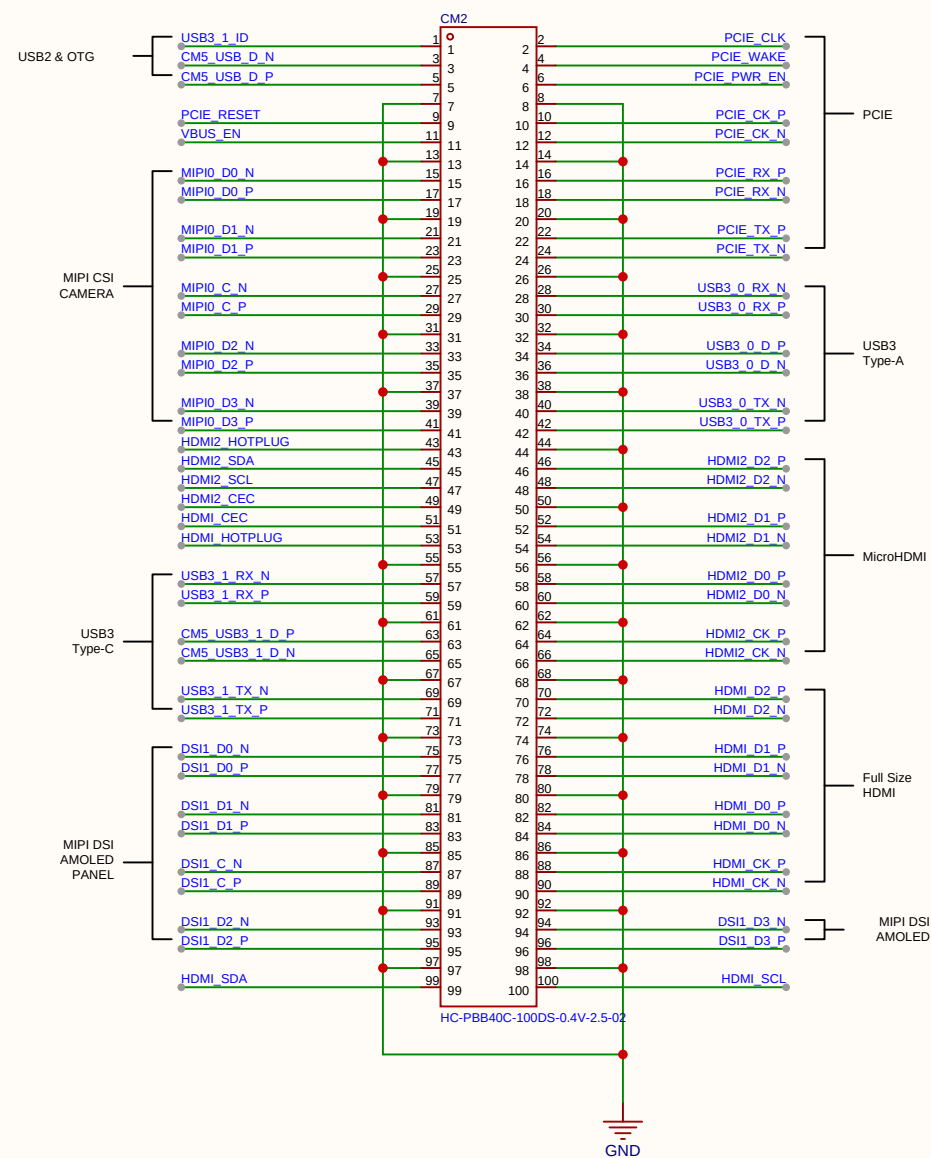


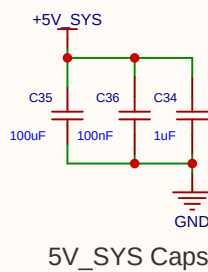
Raspberry Pi CM5 Pinout



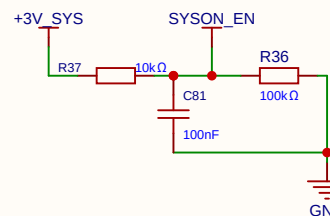
Low-Speed & GPIO Connector (Bottom)



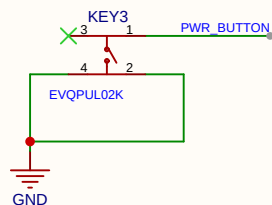
High-Speed Connector (TOP)



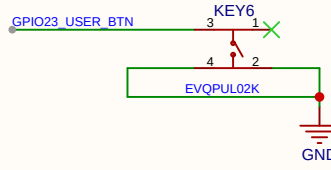
5V_SYS Caps



System ON Signals for Power Switches



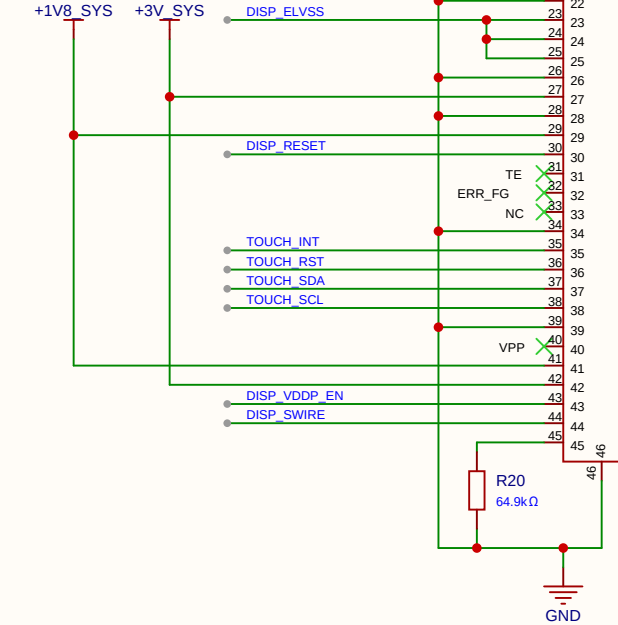
CM5 Power Button



CM5 Based User Button
GPIO 23

Schematic	Pocket CM5			Create at	2025-11-06
				Update at	2026-02-24
Board	Mainboard			Page	CM5
Drawn		piBrick Pocket-CM5			
Reviewed					
		Version	Size	Page 1 Total 8	
		V1.0	A3	amarullz.com	

3.91 Inch Amoled 1080x1240@60Hz - OnCell Multitouch - VTDR6110
0.3mm Pitch 45pin FPC Connector - Bottom Contact



PCB Layout Reference:
https://www.richtek.com/assets/product_file/RT4722C/DS4722C-00.pdf

HDMI2 5V

HDMI2

1 HOT_PLUG_DET

2 UTILITY

3 TMDS_DATA2+

4 TMDS_DATA2_SHIELD

5 TMDS_DATA2-

6 TMDS_DATA1+

7 TMDS_DATA1_SHIELD

8 TMDS_DATA1-

9 TMDS_DATA0+

10 TMDS_DATA0_SHIELD

11 TMDS_DATA0-

12 TMDS_CLOCK+

13 TMDS_CLOCK_SHIELD

14 TMDS_CLOCK-

15 CEC

16 DDC/CEC_GND

17 SCL

18 SDA

19 +5V

20 SHIELD

HDMI101W

HDMI2_HOTPLUG

HDMI2_D2_P

HDMI2_D2_N

HDMI2_D1_P

HDMI2_D1_N

HDMI2_D0_P

HDMI2_D0_N

HDMI2_CK_P

HDMI2_CK_N

HDMI2_CEC

HDMI2_SCL

HDMI2_SDA

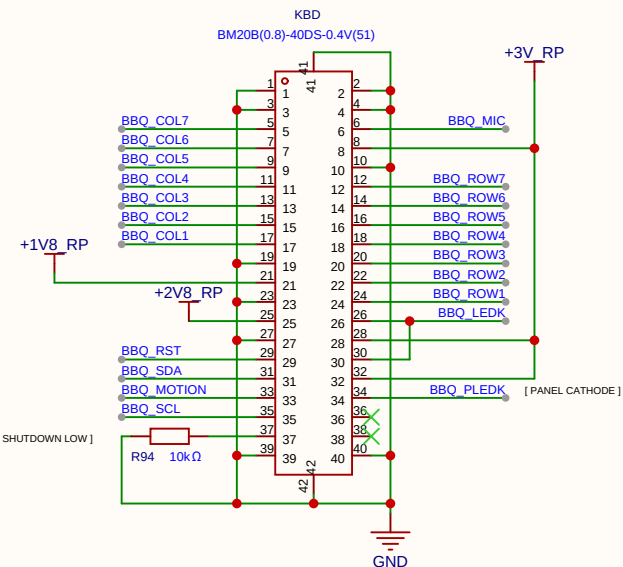
TVS6

C20

100nF

GND

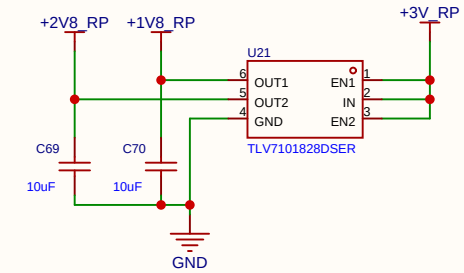
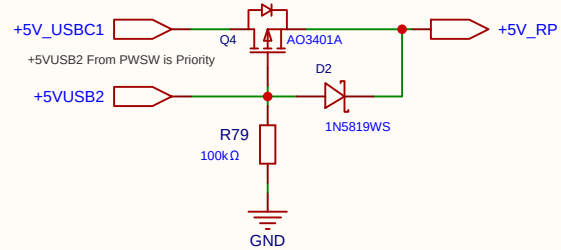




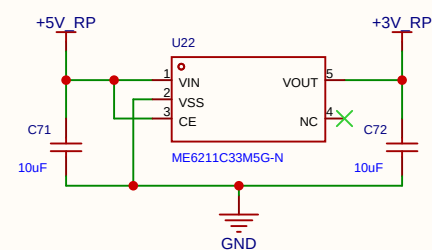
BBQ20 Keyboard Connector

HID Power Or-ing

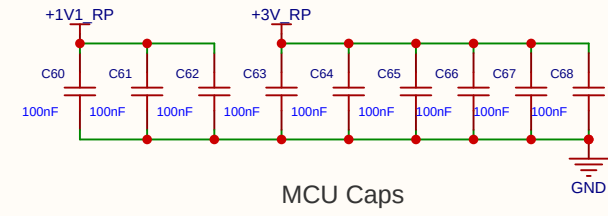
- Power from bottom USB-C when SYS is Off
- 3v3, 1v8 + 2v8 for BBQ20 Keyboard



+1V8 & +2V8 Regulator

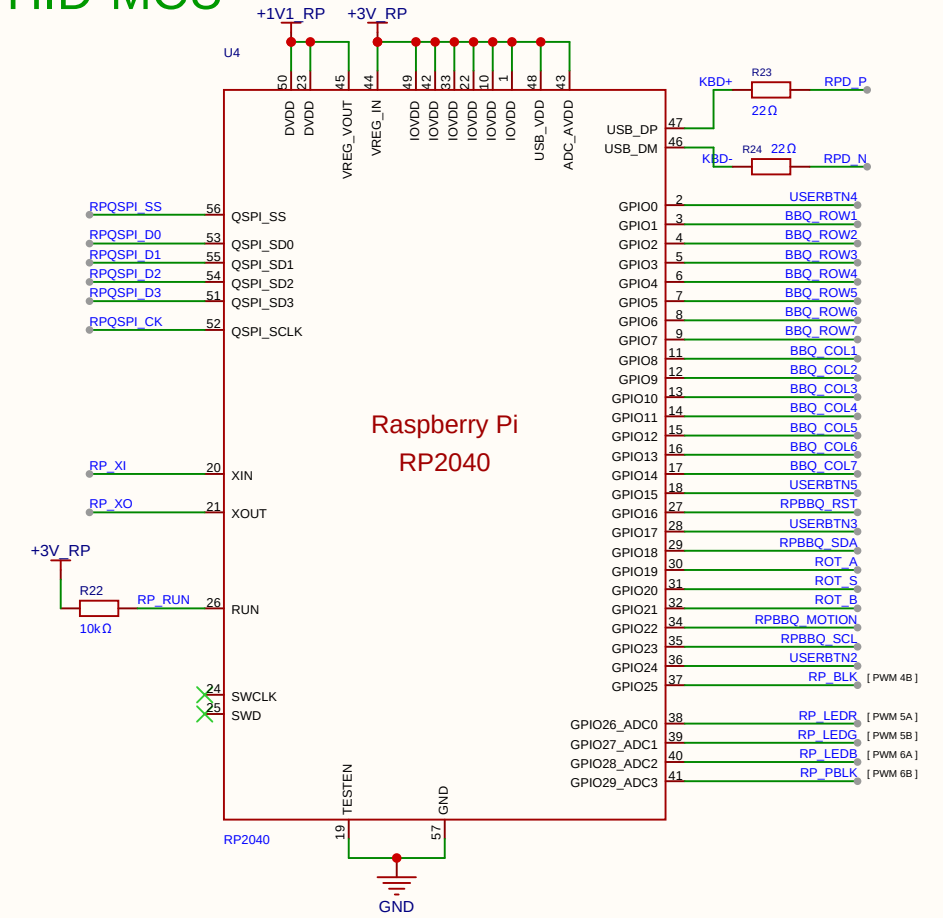


+3V3 Regulator

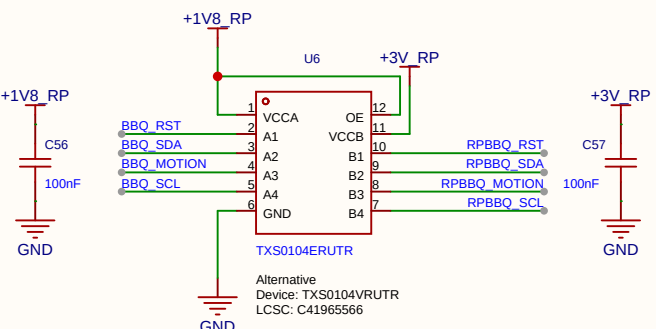


MCU Caps

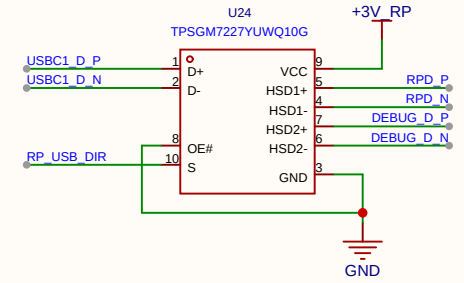
RP2040 HID-MCU



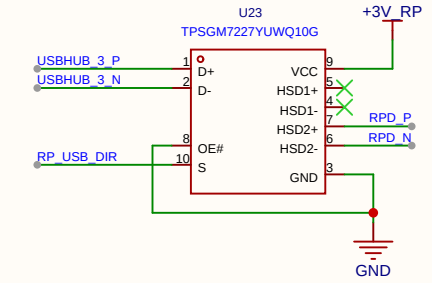
Raspberry Pi
RP2040



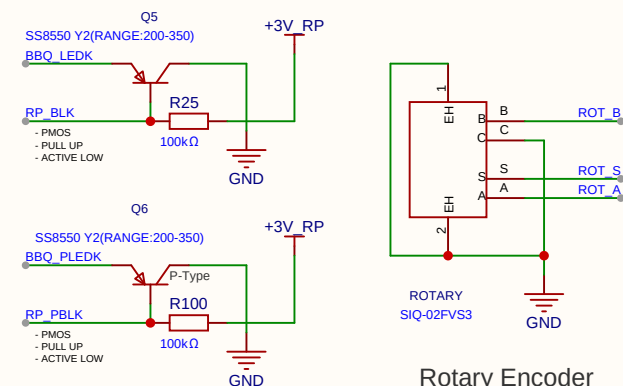
Trackpad Level Shifter



HID/DEBUG to Type-C MUX

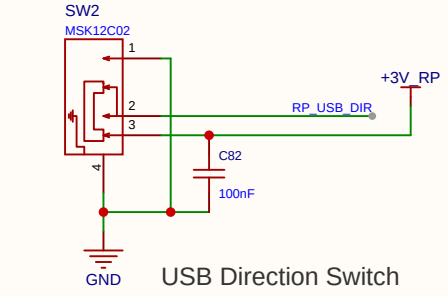


HID to HUB->CM5 MUX

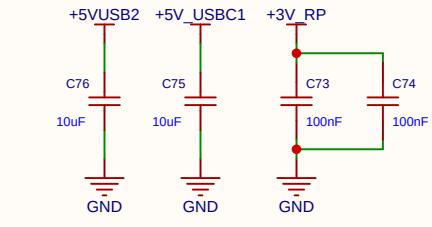


Backlight Mosfet

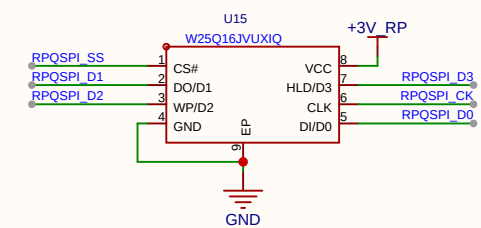
Rotary Encoder



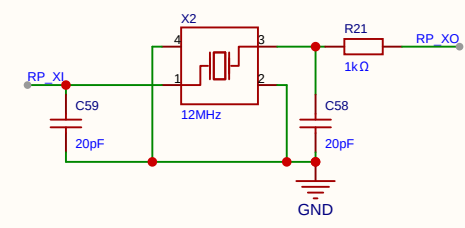
USB Direction Switch



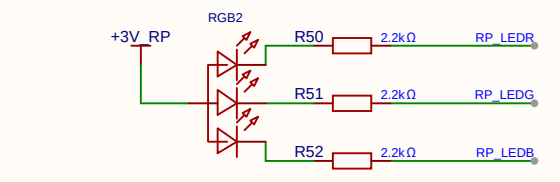
Power Switch Caps



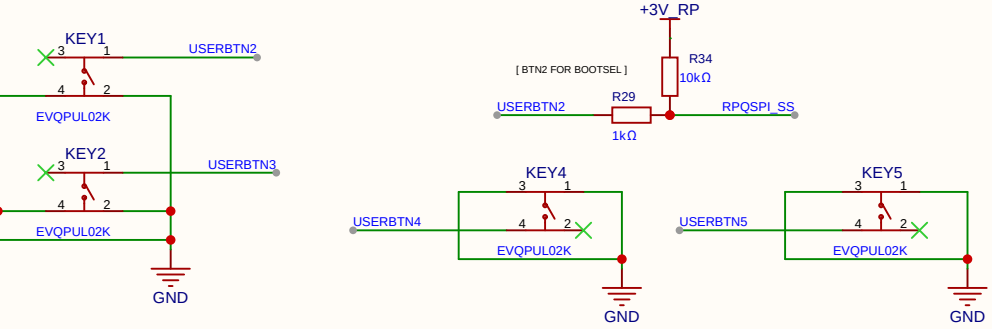
RP2040 Flash



RP2040 Clock



RGB LED for Layer Layout Indicator

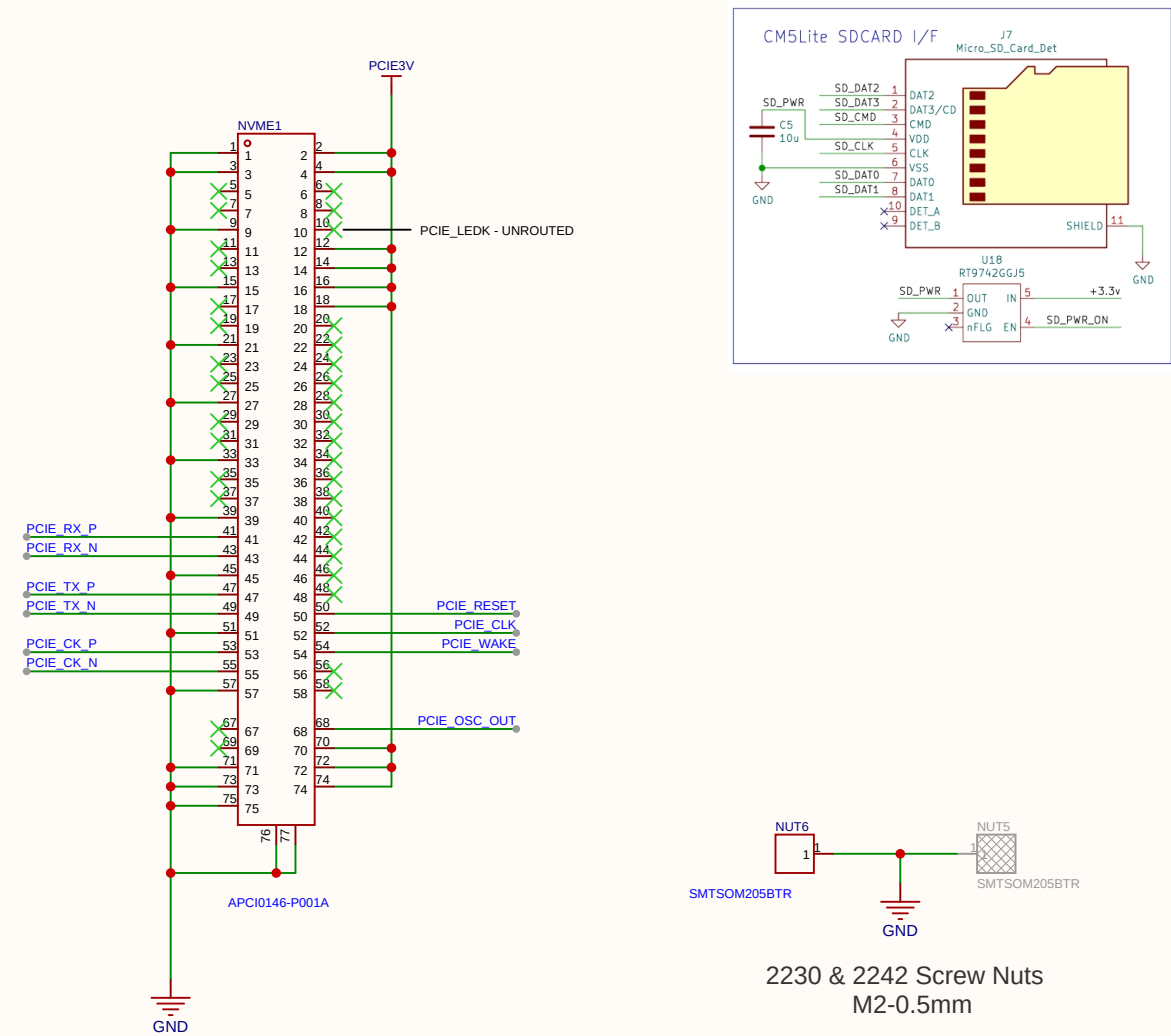


User Buttons & RP2040 Boot Button

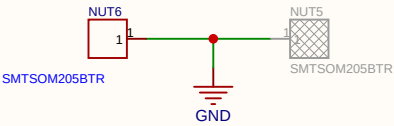
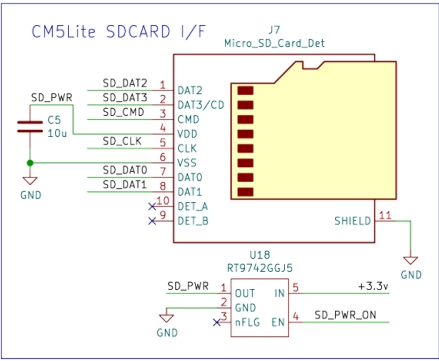
Schematic	Pocket CM5			Create at	2025-11-06
				Update at	2026-02-12
	Board	Mainboard			Page
Drawn	piBrick Pocket-CM5			HID	
Reviewed					
		Version	Size	Page 4 Total 8	
		V1.0	A3	amarullz.com	

PCIE M.2 NVME Connector

1x PCIE 2/3 - 3V 3A Power - 2.5mm Connector Height | Support 2230 & 2242

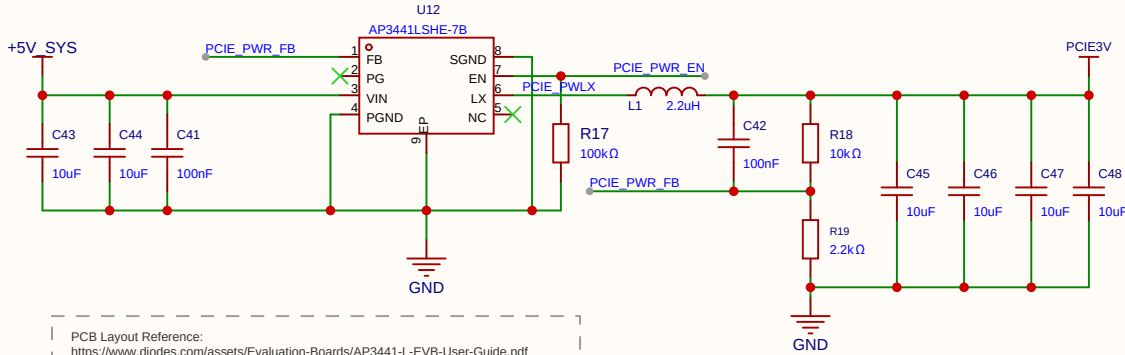


M.2 NVME Connector



2230 & 2242 Screw Nuts
M2-0.5mm

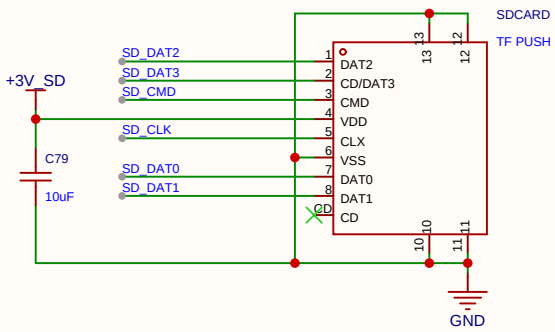
DC to DC Alternative:
- AP3441SHE-7B (C2070918)
- AP3441LSHE-7B (C2072094)



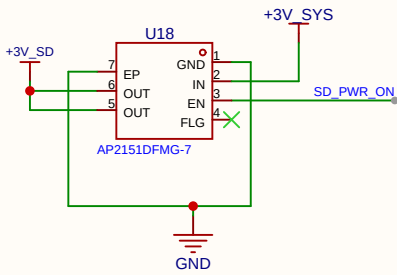
PCB Layout Reference:
<https://www.diodes.com/assets/Evaluation-Boards/AP3441-L-EVB-User-Guide.pdf>

PCIE POWER REGULATOR

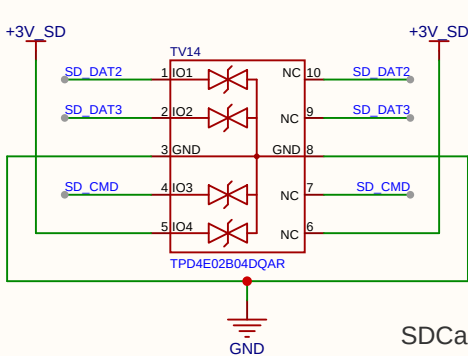
SDCard



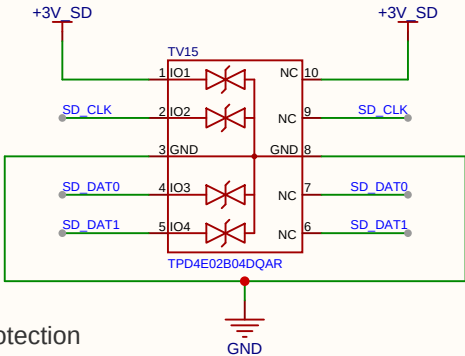
SDCard Push-Push Connector



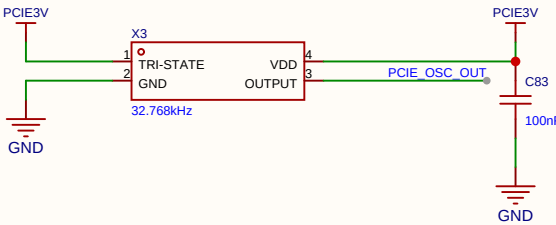
SDCard Power Switch



SDCard ESD Protection



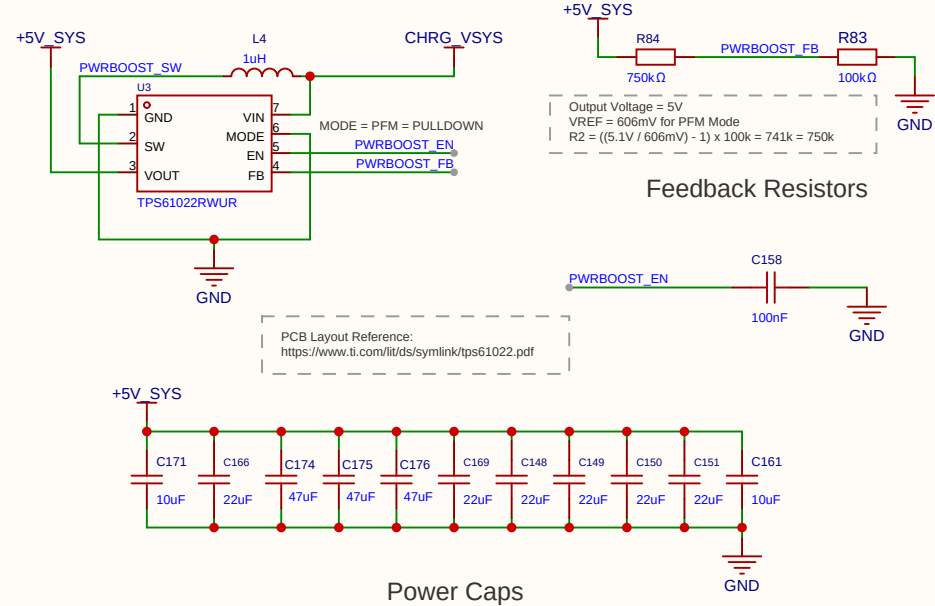
PCIE 32.768kHz OSC



Schematic	Pocket CM5			Create at	2025-11-06
				Update at	2026-03-17
Board	Mainboard			Page	STORAGES
Drawn		piBrick Pocket-CM5			
Reviewed					
		Version	Size	Page 5 Total 8	
 amarullz.com open source hardware		V1.0	A3	amarullz.com	

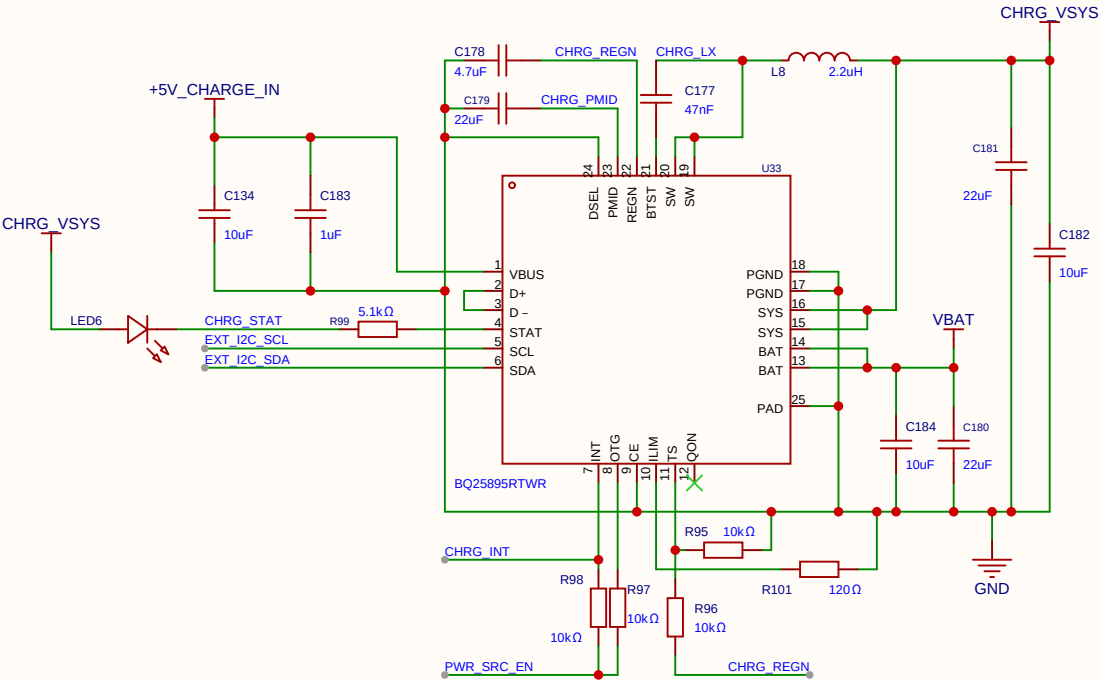
5V Power - Boost

- TPS61022 Boost IC - 5V Max 3A



Battery Charger

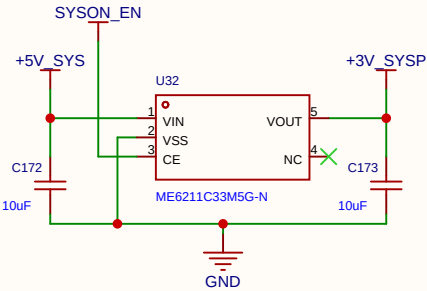
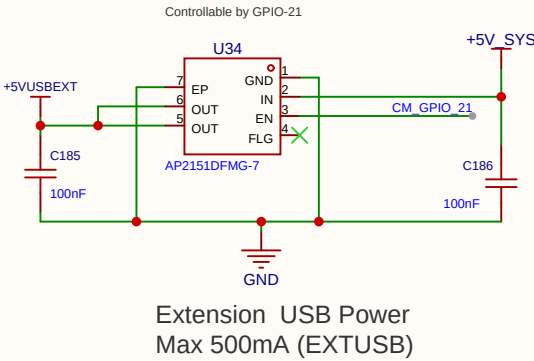
- TI BQ25895RTWR
- Charger LED Indicator



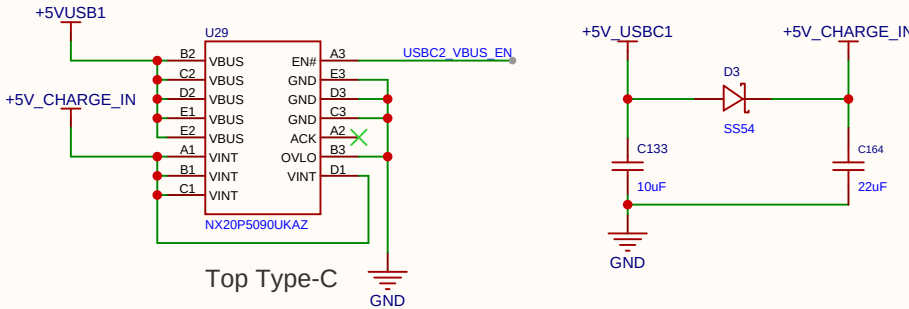
@Credit to: azlan.works

3V SYS Peripherals

- 500mA 3.3V Power
- Powered when CM5 on, same with 3V_SYS



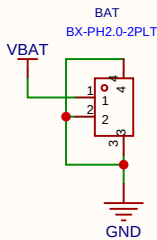
Top Type-C & Bottom Type-C
Or-ing to Input Charge



CHARGING FROM TOP TYPE-C
SHOULD BE MORE EFFICIENT

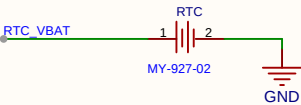
Battery Connector

- JST 2pin 2mm Pitch
- LiPo 3.7V-4.2V 5000mAh (80x50x10mm)

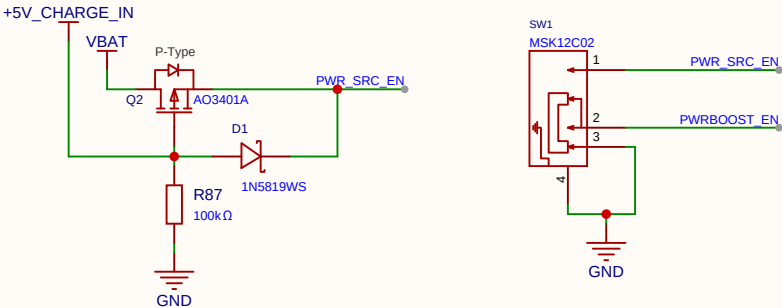


RTC Battery Connector

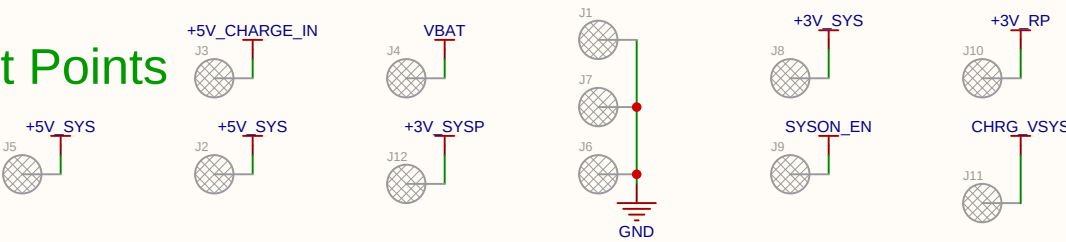
- Coin Battery for RTC - CR927 3V



ON/OFF Power Switch



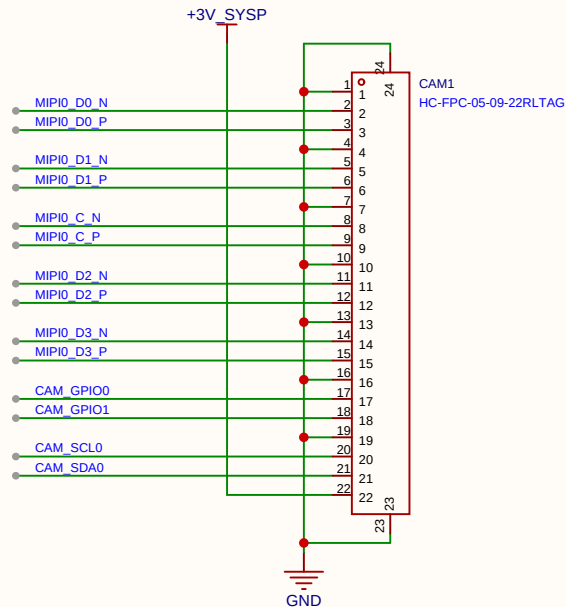
Test Points



Schematic	Pocket CM5		Create at	2025-11-06
Board	Mainboard		Update at	2026-04-14
Drawn			Page	POWER
Reviewed			piBrick Pocket-CM5	
		Version	Size	Page 6 Total 8
		V1.0	A3	amarullz.com

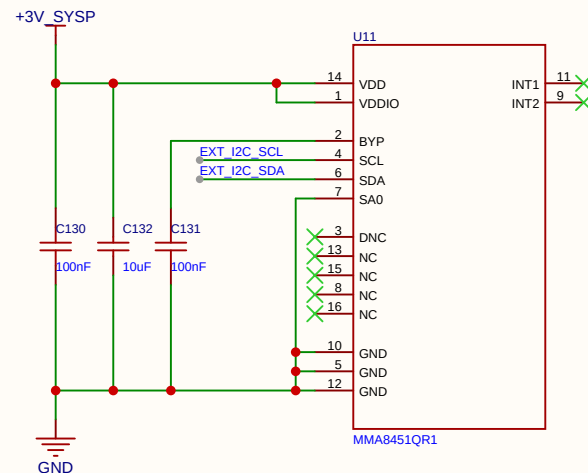
Camera Connector

22pin 0.5mm pitch Standard RPi (Flipped | Top Contact)



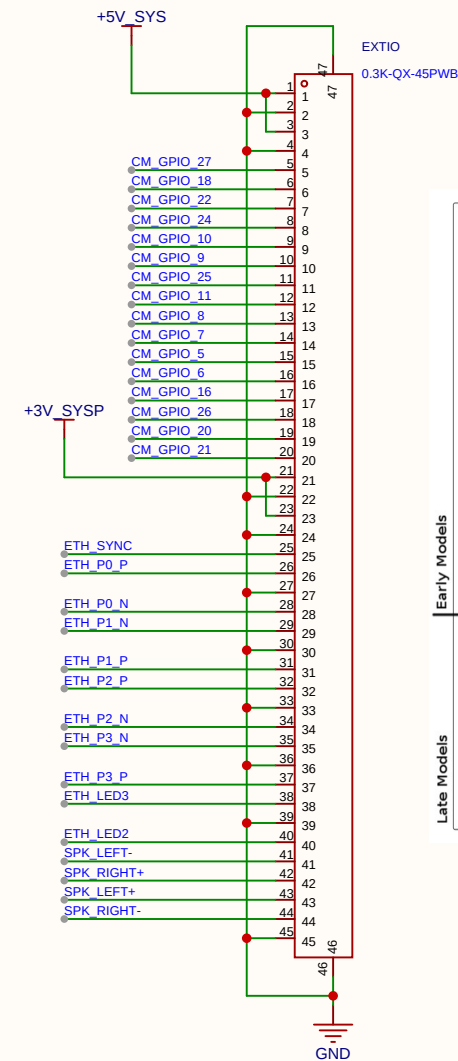
Accelerometer Sensor

- NXP MMA8451QR1
- I2C Address: 0x1C



GPIO, Speaker, Ethernet Extension Connector

45pin 0.35mm pitch (Bottom Contact)

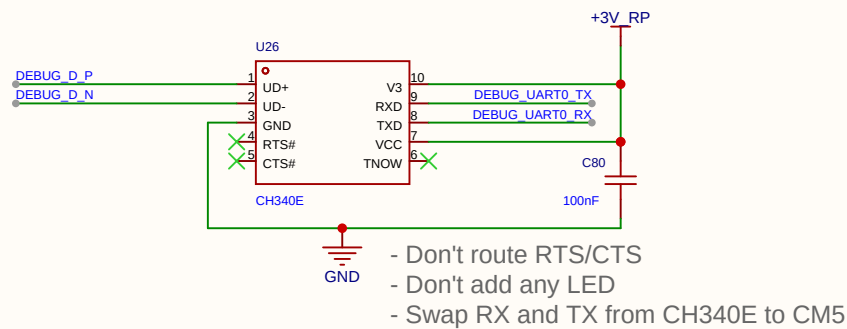


Raspberry Pi2 GPIO Header			
Pin#	NAME	NAME	Pin#
01	3.3v DC Power	DC Power 5v	02
03	GPIO:2 (SDA1 , PC)	DC Power 5v	04
05	GPIO:3 (SCL1 , PC)	Ground	06
07	GPIO:4 (GPIO_GCLK)	(TXD0) GPIO14	08
09	Ground	(RXD0) GPIO15	10
11	GPIO17 (GPIO_GEN0)	(GPIO_GEN1) GPIO18	12
13	GPIO27 (GPIO_GEN2)	Ground	14
15	GPIO22 (GPIO_GEN3)	(GPIO_GEN4) GPIO23	16
17	3.3v DC Power	(GPIO_GEN5) GPIO24	18
19	GPIO10 (SPI_MOSI)	Ground	20
21	GPIO:9 (SPI_MISO)	(GPIO_GEN6) GPIO25	22
23	GPIO11 (SPI_CLK)	(SPI_CE0_N) GPIO:8	24
25	Ground	(SPI_CE1_N) GPIO:7	26
27	ID_SD (PC ID EEPROM)	(PC ID EEPROM) ID_SC	28
29	GPIO:5	Ground	30
31	GPIO:6	GPIO12	32
33	GPIO13	Ground	34
35	GPIO19	GPIO16	36
37	GPIO26	GPIO20	38
39	Ground	GPIO21	40

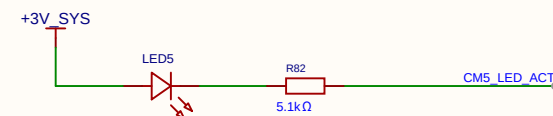
Rev. 1
26/01/2014
<http://www.element14.com>

USB to UART Debug

- CH340E Crystal-less USB to UART
- Port UART0 (RPi GPIO-14 & 15)

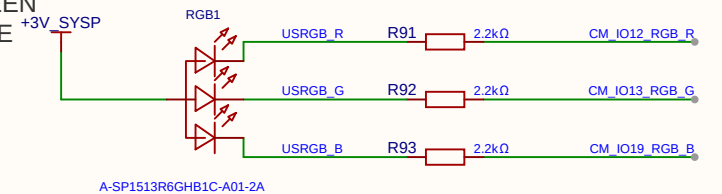


CM5 Activity Indicator



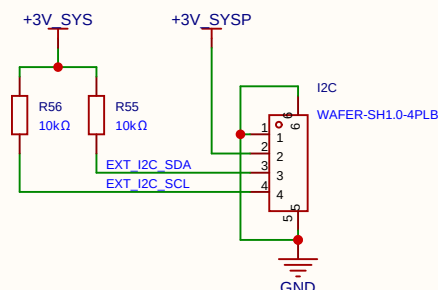
USER RGB Indicator

- Controlled via CM5 GPIO
- PWM channel 0 - GPIO12 - RED
- PWM channel 1 - GPIO13 - GREEN
- PWM channel 1 - GPIO19 - BLUE



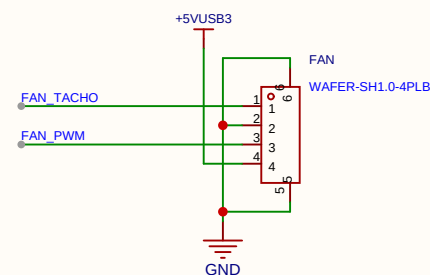
I2C External

- Port I2C1 (RPi GPIO-2 & 3)
- +3V from CM5 directly



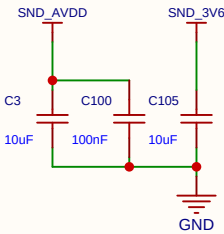
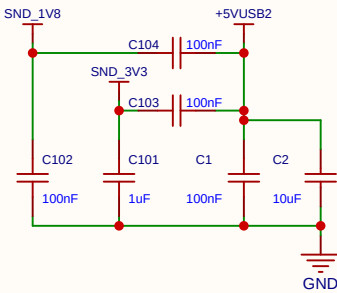
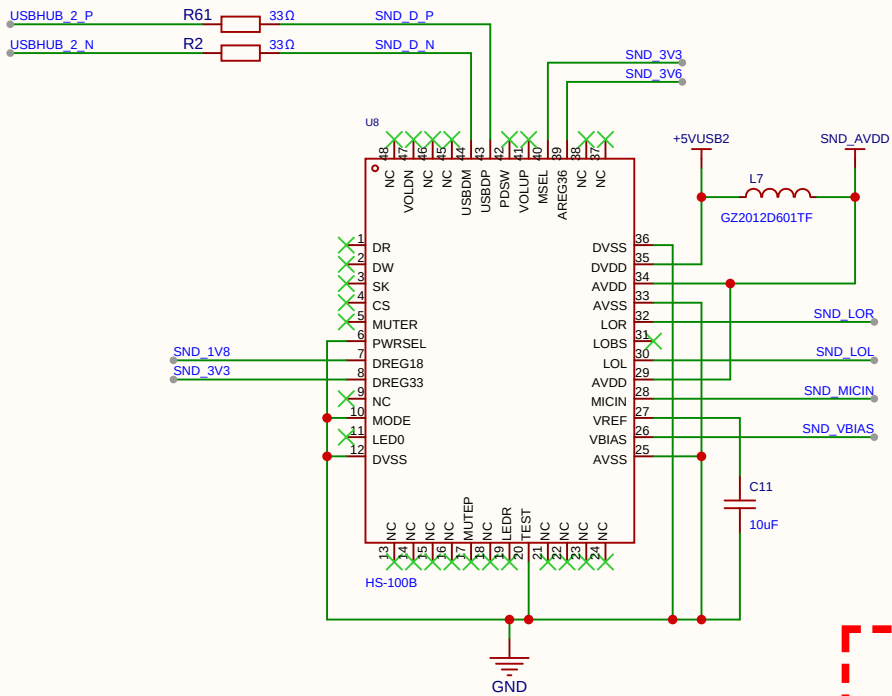
CM5 Heatsink FAN - PWM

- +5V from Power Distributor 3
- PWM & Tacho Signals from CM5
- Only powered if CM5 is ON

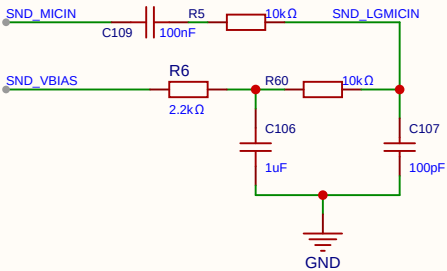


Schematic	Pocket CM5			Create at	2025-11-06
Board	Mainboard			Update at	2025-12-08
Drawn				Page	PERIPHERALS
Reviewed				piBrick Pocket-CM5	
		Version	Size	Page 7 Total 8	
		V1.0	A3	amarullz.com	

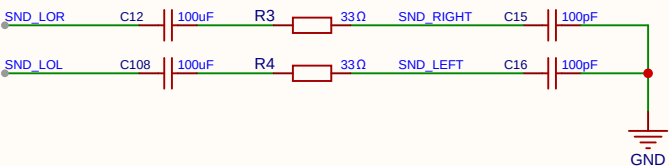
USB Audio - HS-100B



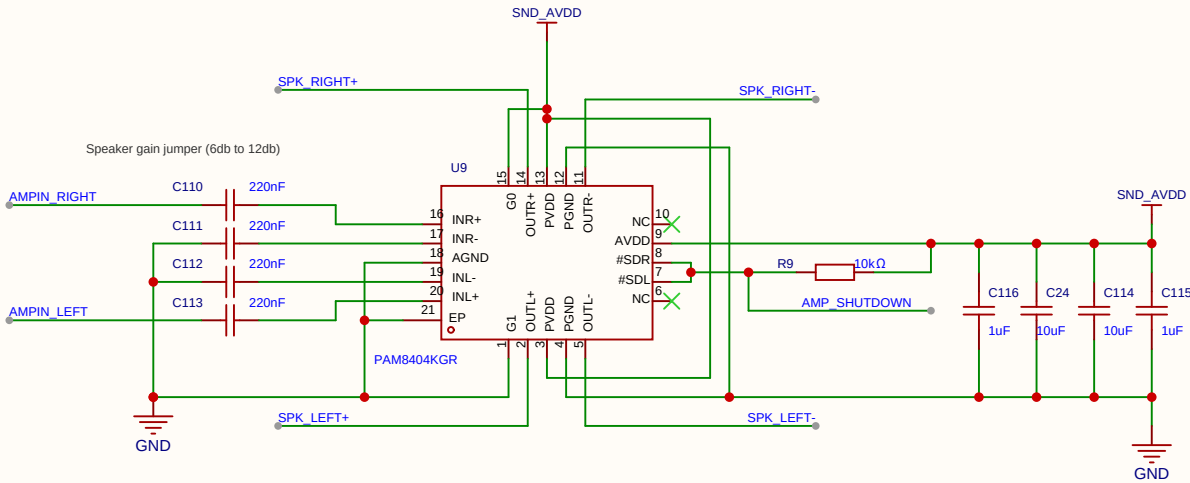
Microphone Input



Low Gain Output

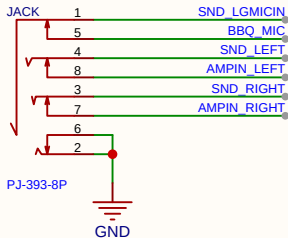


Audio Amplifier - PAM8404KGR

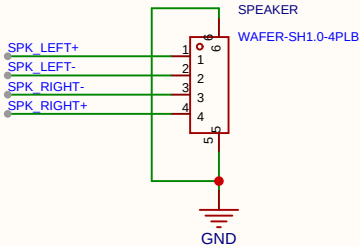


3.5mm Jack

- Auto turn off speaker & BBQ20 Mic on Jack Plug



Speaker



Schematic	Pocket CM5			Create at	2025-11-06
				Update at	2026-01-24
Board	Mainboard			Page	AUDIO
Drawn		piBrick Pocket-CM5			
Reviewed					
		Version	Size	Page 8 Total 8	
		V1.0	A3	amarullz.com	